

Lucas Alves

Data Engineer | Python, SQL, Amazon Redshift | ETL Pipelines & Process Automation
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PROFESSIONAL SUMMARY

Data Engineer with 5+ years building production ETL pipelines processing data from **dozens of Brazilian cities**, automating data collection at scale, and leading cross-functional development teams. Specialist in **Python, SQL, Amazon Redshift, DuckDB, and PostgreSQL**, delivering end-to-end data integration and analytics solutions for urban mobility clients nationwide. Accelerates delivery with AI-assisted development (**Claude, GitHub Copilot**) for pipeline scaffolding, SQL generation, and automated documentation. Published researcher in urban mobility data modeling with graduate coursework in Intelligent Sensing and Geospatial Data. Proven track record: eliminated manual processes cutting team workloads by 2/3, automated data acquisition reducing collection from hours to minutes, and scaled prospecting capacity by 10x through custom tooling. Multiple national programming competition medals (OPI Silver, top 10% Brazilian Programming Olympiad). B.Sc. in Computer Engineering (IFPB, 2025). DataCamp certified in Data Science, Data Analysis, and AI Fundamentals. Native Portuguese and advanced English (B2/C1).

PROFESSIONAL EXPERIENCE

• Project Manager

Colégio Panorama · Campina Grande, Brazil · On-site

Dec 2025 - Present

- Lead multidisciplinary teams responsible for the institution's digital transformation, managing projects spanning data analysis, software development, and process automation using Scrum and Kanban; leverage **AI-assisted development tools** to accelerate delivery and improve code quality across the team.
- Coordinated implementation of a **Python ETL pipeline** (SQLAlchemy → PostgreSQL) that eliminated manual data transfers consuming **2/3 of the team's weekly hours**; supervised a **JavaScript REST API** generating interactive student performance dashboards — delivering unprecedented analytical visibility to school management.
- Structured version control processes with GitHub, standardizing the team's development workflow and reducing delivery friction across concurrent initiatives.

• Data Analyst / Data Engineer

Bus2 · Brazil · 100% Remote

May 2023 - Present

- Build and optimize **ETL pipelines** integrating heterogeneous data sources from **dozens of Brazilian cities** using Python, Knime, Amazon Redshift, and DuckDB; accelerate pipeline development and SQL generation with **Claude and GitHub Copilot**, reducing development cycles and improving code quality.
- Conduct city-wide boarding/alighting transit research, identifying passenger behavior patterns that directly guide municipal **route planning and scheduling decisions**; deliver actionable insights on population mobility to municipal clients across the country.
- Reduced data acquisition time from **hours to minutes** by replacing manual collection with Python + Selenium web scraping; optimize pipeline processing, reducing bottlenecks and increasing data reliability across electronic ticketing systems for municipal clients nationwide.
- Document data engineering processes ensuring traceability and facilitating solution scalability; provide technical mentorship and knowledge sharing that reduced onboarding time for new analysts and contributed to workflow standardization across the team.

• Software Engineer

Colégio Panorama · Campina Grande, Brazil · On-site

Jun 2021 - May 2023

- Created a **Python ETL pipeline** (SQLAlchemy → PostgreSQL) eliminating manual data transfers that consumed **2/3 of the team's weekly time**; developed a **JavaScript REST API** with SQL queries generating interactive student performance charts — delivering analytical visibility to management for the first time.
- Automated electronic invoice (NFs-e) issuance via WebISS with Python, eliminating **5 full days/month** of manual work with zero errors; administered PostgreSQL database (installation, configuration, monitoring, optimization) ensuring continuous availability and performance.
- Implemented web scraping solutions with Python and Selenium for automated periodic data collection from external sources, replacing manual processes entirely.

• Software Engineer

Multi Telecom · Campina Grande, Brazil · On-site

Aug 2020 - Jun 2021

- Developed **Python bots** (Selenium + PyQt5) for WhatsApp automation, multiplying the sales team's prospecting capacity by **10x**; built a phone number validation tool processing **10K+ numbers per batch**, exporting results to structured spreadsheets.
- Built automated file handling system (PyQt5 + SQLite3 + Pandas) for spreadsheet manipulation and local network distribution, freeing up **20 hours/week** of team time and eliminating manual transfer errors.
- Created a **JavaScript chatbot** (wa-automate-js) for WhatsApp customer service, automating frequent responses and significantly reducing response time.

TECHNICAL SKILLS

Languages: Python (advanced), JavaScript, SQL, HTML, CSS.

AI & Automation: Claude, GitHub Copilot, AI-assisted pipeline scaffolding, SQL generation, automated documentation, prompt engineering.

Data & ETL: Pandas, SQLAlchemy, Selenium, Knime, DuckDB, ETL/ELT pipeline design, web scraping, data quality monitoring.

Databases & Cloud: PostgreSQL, Amazon Redshift (data warehousing, complex queries), SQLite, DuckDB; AWS, Docker, Linux.

Backend & Frameworks: Django, FastAPI, Node.js, REST APIs; Git, GitHub, GitHub Actions, CI/CD.

Analytics & Visualization: Plotly, data modeling, statistical analysis, report generation, dashboard development.

Management & Methodology: Scrum, Kanban, cross-functional team leadership, full project lifecycle management, technical mentorship.

PUBLICATIONS & PROJECTS

- "Performance analysis of transportation systems through the boarding/alighting method" — peer-reviewed research
- "Modeling urban mobility patterns: boarding and alighting analysis in transit systems in Campina Grande" — peer-reviewed research
- Google Maps Crawler** — Python, Selenium, xlwings, PyQt5. Automated business data extraction from Google Maps to Excel, 1K+ records/run with configurable search parameters.

ACHIEVEMENTS

- Silver Medal — Olimpíada Paraibana de Informática (OPI), Advanced Senior Category 2022
- Top 10% — XXII Brazilian Programming Olympiad 2022
- 3rd Place — Maratona POP de Programação, Advanced Category 2022
- 5th Place — Maratona POP de Programação, Beginner Category 2021
- 4th Place — Maratona POP de Programação, Beginner Category 2020

CERTIFICATIONS

- Data Scientist Associate — DataCamp
- Data Analyst Associate — DataCamp
- AI Fundamentals — DataCamp
- Introduction to Data Visualization with Plotly in Python — DataCamp
- Analyzing Police Activity with pandas — DataCamp

EDUCATION

• M.Sc. (Special Student) — Intelligent Sensing and Geospatial Data

Instituto Federal da Paraíba (IFPB), Campina Grande

Aug 2025 - Dec 2025

- Emphasis on geospatial data processing, intelligent sensing systems, and applied data science.

• Bachelor's in Computer Engineering

Instituto Federal da Paraíba (IFPB), Campina Grande

2020 - 2025

- Published researcher in urban mobility data modeling. Multiple competitive programming awards (OPI Silver, top 10% national olympiad).

• Technical Degree in Informatics

Instituto Federal da Paraíba (IFPB), Campina Grande

2017 - 2019

- Foundation in programming, databases, and software development fundamentals.